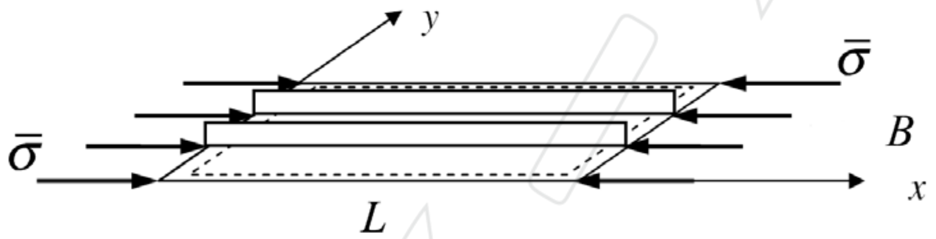
Imperial College London
AE3-304 Aircraft Structures – Plate Bending & Buckling

Tutorial 4

Questions

Consider a rectangular plate *B* wide, *L* long and *t* thick that belongs to the upper skin of a wing. The plate is simply supported all around its contour. At the design limit load, the plate is subject to a uniform longitudinal compressive stress of magnitude $\bar{\sigma}$. In order to prevent buckling, the plate is reinforced with *N* flat longitudinal stringers having height h_s and thickness t_s. You may assume that both the skin and the stringers are made of the same structural Aluminium alloy and that the stringers are evenly spaced with respect to the plate width *B*.



1) Give a conservative estimate of the buckling stress of the plate skin as a function of the number of stringers;

[15 marks]

2) Provide a conservative estimate of the buckling stress of the individual stringers;

[15 marks]

3) Considering airworthiness requirements, express the ratio between the skin thickness and the total skin width *B* with respect to the applied compressive load.

[15 marks]

4) Considering airworthiness requirements, express the ratio between the stringer thickness and the stringer height as a function of the applied compressive load.

[15 marks]

5) The "equivalent thickness" of the plate is defined as follows

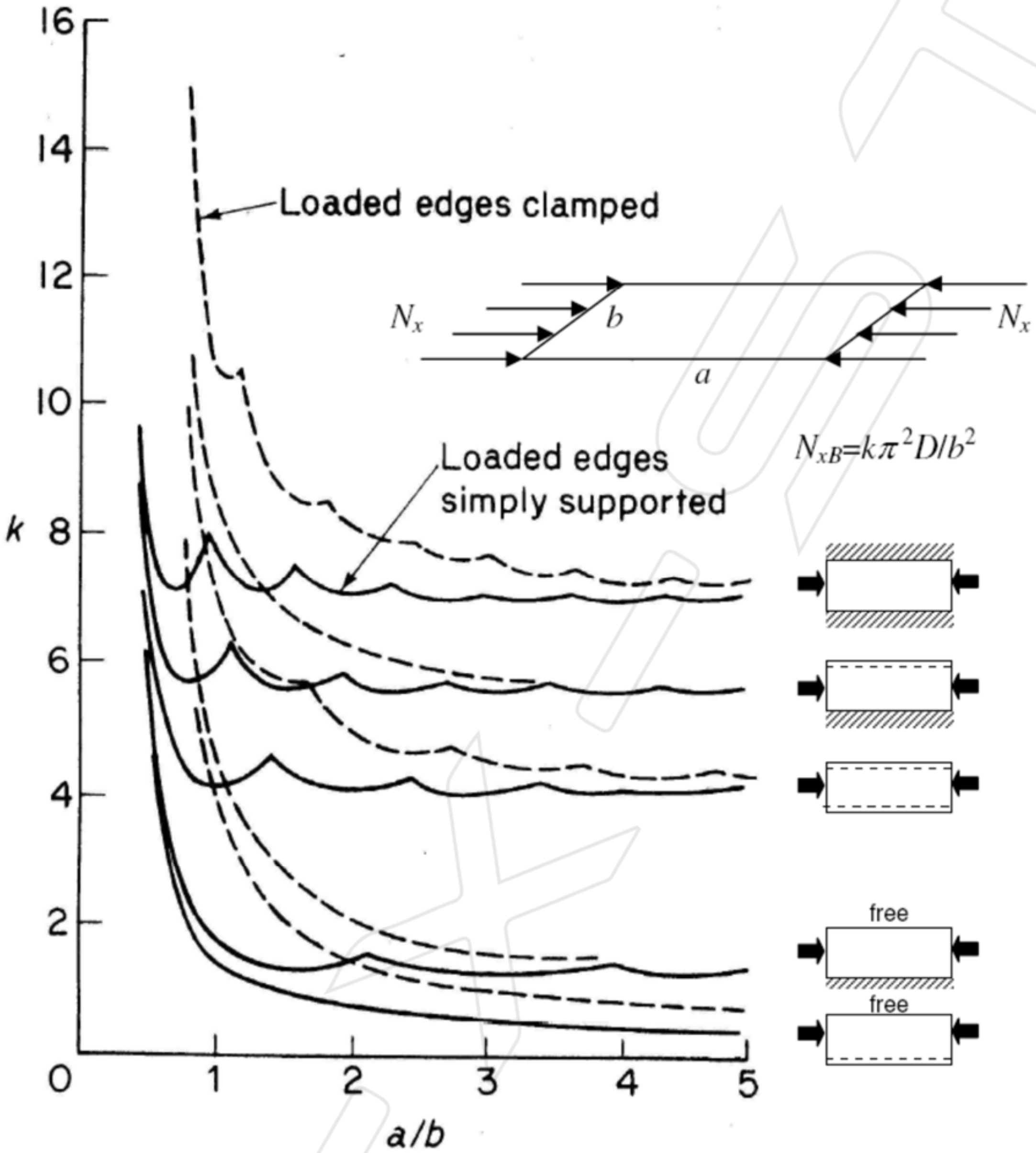
$$t_{eq} = \frac{m_p}{\rho L B}$$

where m_p is the total mass of the plate and ρ the density of the Aluminium alloy employed. Express the equivalent thickness as a function of the number of stringers, stringer height and pitch.

[25 marks]

6) Plot the equivalent thickness with respect to the number of stringers and discuss if and how the mass of the plate can be minimised while satisfying the buckling constraints.

[15 marks]


Loaded edges clamped
Loaded edges simply supported
N_x
b
a
N_x
$N_{xB}=k\pi^2D/b^2$
free
free
k
a/b
0
1
2
3
4
5
2
4
6
8
10
12
14
16

for simply-supported boundary condition, $w = A_{mn} \sin\frac{m\pi x}{a} \sin\frac{n\pi y}{b}$

$$\therefore \frac{\partial w}{\partial x} = A_{mn} \frac{m\pi}{a} \cos\frac{m\pi x}{a} \sin\frac{n\pi y}{b}$$

$$\frac{\partial w}{\partial y} = A_{mn} \frac{n\pi}{b} \sin\frac{m\pi x}{a} \cos\frac{n\pi y}{b}$$

$$\therefore \frac{\partial^2 w}{\partial x^2} = -A_{mn} \left(\frac{m\pi}{a}\right)^2 \sin\frac{m\pi x}{a} \sin\frac{n\pi y}{b}$$

$$\frac{\partial^2 w}{\partial y^2} = -A_{mn} \left(\frac{n\pi}{b}\right)^2 \sin\frac{m\pi x}{a} \sin\frac{n\pi y}{b}$$

$$\frac{\partial^2 w}{\partial x \partial y} = A_{mn} \left(\frac{m\pi}{a}\right)\left(\frac{n\pi}{b}\right) \cos\frac{m\pi x}{a} \cos\frac{n\pi y}{b}$$

$$\therefore \left(\frac{\partial w}{\partial x}\right)^2 = A_{mn}^2 \left(\frac{m\pi}{a}\right)^2 \cos^2\frac{m\pi x}{a} \sin^2\frac{n\pi y}{b}$$

$$\left(\frac{\partial^2 w}{\partial x^2}\right)^2 = A_{mn}^2 \left(\frac{m\pi}{a}\right)^4 \sin^2\frac{m\pi x}{a} \sin^2\frac{n\pi y}{b}$$

$$\left(\frac{\partial^2 w}{\partial y^2}\right)^2 = A_{mn}^2 \left(\frac{n\pi}{b}\right)^4 \sin^2\frac{m\pi x}{a} \sin^2\frac{n\pi y}{b}$$

$$\frac{\partial^2 w}{\partial x^2}\frac{\partial^2 w}{\partial y^2} = A_{mn}^2 \left(\frac{m\pi}{a}\right)^2\left(\frac{n\pi}{b}\right)^2 \sin^2\frac{m\pi x}{a} \sin^2\frac{n\pi y}{b}$$

$$\left(\frac{\partial^2 w}{\partial x \partial y}\right)^2 = A_{mn}^2 \left(\frac{m\pi}{a}\right)^2\left(\frac{n\pi}{b}\right)^2 \cos^2\frac{m\pi x}{a} \cos^2\frac{n\pi y}{b}$$

$$U = \frac{D}{2}\int_0^a\int_0^b \left[\left(\frac{\partial^2 w}{\partial x^2}\right)^2 + \left(\frac{\partial^2 w}{\partial y^2}\right)^2 + 2\nu\left(\frac{\partial^2 w}{\partial x^2}\right)\left(\frac{\partial^2 w}{\partial y^2}\right) + 2(1-\nu)\left(\frac{\partial^2 w}{\partial x \partial y}\right)\right] dx\,dy$$

where $\int_0^a\int_0^b \left(\frac{\partial^2 w}{\partial x^2}\right)^2 dx\,dy = A_{mn}^2 \left(\frac{m\pi}{a}\right)^4\left(\frac{a}{2}\right)\left(\frac{b}{2}\right)$

$$\int_0^a\int_0^b \left(\frac{\partial^2 w}{\partial y^2}\right)^2 dx\,dy = A_{mn}^2 \left(\frac{n\pi}{b}\right)^4\left(\frac{a}{2}\right)\left(\frac{b}{2}\right)$$

$$\int_0^a\int_0^b \frac{\partial^2 w}{\partial x^2}\frac{\partial^2 w}{\partial y^2} dx\,dy = A_{mn}^2 \left(\frac{m\pi}{a}\right)^2\left(\frac{n\pi}{b}\right)^2\left(\frac{a}{2}\right)\left(\frac{b}{2}\right)$$

$$\int_0^a\int_0^b \left(\frac{\partial^2 w}{\partial x \partial y}\right)^2 dx\,dy = A_{mn}^2 \left(\frac{m\pi}{a}\right)^2\left(\frac{n\pi}{b}\right)^2\left(\frac{a}{2}\right)\left(\frac{b}{2}\right)$$

$$\therefore U = \frac{1}{2} D A_{mn}^2 \left(\frac{a}{2}\right)\left(\frac{b}{2}\right)\left[\left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2\right]^2$$

$$W_{op} = 0$$

$$W_{ip} = \frac{1}{2}\int_0^a\int_0^b N_x \left(\frac{dw}{dx}\right)^2 dx\,dy = \frac{1}{2} N_x A_{mn}^2 \left(\frac{m\pi}{a}\right)^2\left(\frac{a}{2}\right)\left(\frac{b}{2}\right)$$

$$\therefore \Pi = U - W = U - (W_{op} + W_{ip})$$

$$= \frac{1}{2} A_{mn}^2 \pi^2 \left\{D\pi^2\left[\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2\right]^2 - N_x \left(\frac{m}{a}\right)^2\right]$$

for $\frac{d\Pi}{dA_{mn}} = 0$ at N_{xB},

$$N_{xB} = D\pi^2 \left(\frac{a}{m}\right)^2 \left[\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2\right]^2$$

for $n = 1$,

$$N_{xB} = D\pi^2 \left(\frac{a}{m}\right)^2 \left[\left(\frac{m}{a}\right)^2 + \left(\frac{1}{b}\right)^2\right]^2 = \frac{D\pi^2}{b^2}\left(\frac{mb}{a} + \frac{a}{mb}\right)^2$$

1) $N_{xB} = \frac{\pi^2 D}{b^2}\left(\frac{mb}{a} + \frac{a}{mb}\right)^2 = k\frac{\pi^2 D}{b^2} = k\frac{\pi^2}{b^2}\frac{Et^3}{12(1-\nu^2)}$

$\therefore \sigma_{xB} = \frac{N_{xB}}{t} = k\frac{\pi^2}{b^2}\frac{Et^2}{12(1-\nu^2)} = KE\left(\frac{t}{b}\right)^2$ where $K = \frac{k\pi^2}{12(1-\nu^2)}$

for conservative estimate, assume the skin between two stiffeners is treated as a simply supported plate with 4 simply supported edges. $\frac{a}{b} > 3$.

$k = 4$ according to the figure above

$\therefore K = 3.62$ for $\nu = 0.3$

$\sigma_{xB,\,skin} = 3.62E\left(\frac{t}{\frac{1}{N+1}B}\right)^2 = 3.62E\left[\frac{(N+1)t}{B}\right]^2$

2) for conservative approach, assume all loaded edge of the stiffeners are simply supported edges and the unloaded edge is free edge. $\frac{a}{b} > 3$

$\therefore K = 0.385$

$\therefore \sigma_{xB,\,stiffener} = 0.385E\left(\frac{t_s}{h_s}\right)^2$

3) considering airworthiness requirements, skin may buckles at the design limit load, $\bar{\sigma}$, therefore

$\bar{\sigma} = 3.62E\left[\frac{(N+1)t}{B}\right]^2$

$\therefore \frac{t}{B} = \frac{1}{N+1}\sqrt{\frac{\bar{\sigma}}{3.62E}}$

4) considering airworthiness requirements, stringers may buckles at the ultimate design load, $\sigma_{ultimate} = 1.5\bar{\sigma}$, therefore

$\sigma_{ultimate} = 1.5\bar{\sigma} = 0.385E\left(\frac{t_s}{h_s}\right)^2$

$\therefore \frac{t_s}{h_s} = \frac{10\sqrt{231}}{77}\sqrt{\frac{\bar{\sigma}}{E}}$

5) $m_p = m_{skin} + N m_{stringer} = \rho BLt + N\rho h_s L t_s$

$\therefore t_{eq} = \frac{\rho BLt + N\rho h_s L t_s}{\rho LB} = t + \frac{N h_s t_s}{B}$

$= \frac{B}{N+1}\sqrt{\frac{\bar{\sigma}}{3.62E}} + \frac{N h_s^2}{B}\frac{10\sqrt{231}}{77}\sqrt{\frac{\bar{\sigma}}{E}}$

$= \left(\frac{B}{N+1}\sqrt{\frac{1}{3.62}} + \frac{N h_s^2}{B}\frac{10\sqrt{231}}{77}\right)\sqrt{\frac{\bar{\sigma}}{E}}$

for stringer pith $b = \frac{B}{N+1}$

$t_{eq} = \left[\sqrt{\frac{50}{181}}\,b + N(N+1)\frac{10\sqrt{231}}{77}\frac{h_s^2}{b}\right]\sqrt{\frac{\bar{\sigma}}{E}}$

6) see solution